Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In step-deviation method, $u_i = \dfrac{x_i - a}{h}$, where h is assumed to be: [1]

(A) any real number (B) common factor of class marks
(C) common class size of the intervals (D) total frequency

Q2. If the mean of first n natural numbers is $\dfrac{5n}{9}$, then the value of n is: [1]

(A) 5 (B) 7 (C) 9 (D) 10

Q3. For a distribution, if $\sum f_i = 100$, assumed mean $a = 25$, $\sum f_i d_i = 20$, then the mean is: [1]

(A) 25.2 (B) 24.8 (C) 27.0 (D) 25.0

Q4. Which of the following measures of central tendency is least affected by extreme values (outliers)? [1]

(A) Arithmetic Mean (B) Median (C) Mode (D) Range

Q5. **Assertion (A):** The median of a continuous grouped distribution can be computed without altering open-ended outermost classes, provided the median class itself is not an open-ended class.

Reason (R): In the formula Median $= l + \left(\dfrac{N/2 - cf}{f}\right) h$, the value depends strictly on the median class boundaries, half the total frequency $(N/2)$, and the preceding cumulative frequency (cf). [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. If the median of a distribution is 26 and its mean is 24, find the mode using the empirical relationship. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the mode of the ages of 80 patients admitted to a clinic: [3]

Age (years)	10–20	20–30	30–40	40–50	50–60	60–70
Patients	12	18	22	16	8	4

Q8. If the mean of the following frequency distribution is 54, find the value of p: [3]

Class Interval	0–20	20–40	40–60	60–80	80–100
Frequency	7	p	10	9	13

Q9. Find the median marks from the following cumulative distribution of 80 students: [3]

Marks	< 10	< 20	< 30	< 40	< 50	< 60
Students (cf)	7	17	37	57	75	80

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: ANNUAL INTER-SCHOOL ATHLETICS 100M SPRINT TIMING ANALYSIS [4 Marks]

During an inter-district track and field meet, the race completion timings of 40 sprinters in the 100m heats were recorded using automated digital laser timing gates.

Time (seconds)	13.5–14.0	14.0–14.5	14.5–15.0	15.0–15.5	15.5–16.0
Number of Sprinters	4	8	15	11	2

(i) Find the class mark and width of the modal sprint timing class. [1]

(ii) Calculate the median race completion time of the athletes. [2]

(iii) How many sprinters finished the 100m dash in under 15.0 seconds? [1]

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